

Jane Street Puzzle - March 2026

Planetary Parade

We are enjoying a planetary parade during the beginning of this month here on Earth, but my alien friend from the distant planet Pyrknor (well modeled as a perfect sphere) isn't too impressed. Their single-star system has six planets other than Pyrknor with such chaotic orbits that each night these planets independently appear in uniformly random locations in the Pyrknor sky (the planets aren't visible in daylight). If at a given moment there exists somewhere on Pyrknor that all six planets are visible, then from my friend's position on the surface they have an α probability of also seeing all of the planets.

My friend is considering building a tower to improve their chances of seeing these planetary parades. Assume the tower allows the viewing of every celestial body that could be seen from at least one location on the surface of Pyrknor less than a distance r from the base of the tower. In the limit of r being small compared to the radius R of Pyrknor, the new probability of seeing the planetary parade from the top of the tower is linearly approximated by $\alpha + \beta(r/R)$. Find α and β in exact terms.

Computing α

Let the positions of the 6 planets be represented as 6 independent, uniformly distributed random points on a celestial sphere.

- Let E be the event that all 6 planets are visible from *somewhere* on Pyrknor (i.e., they all lie within some hemisphere).
- Let A be the event that all 6 planets are visible from the observer's *specific* location (i.e., they lie within the observer's predefined hemisphere).

The unconditional probability that all 6 planets fall into the observer's specific hemisphere is simply:

$$P(A) = \left(\frac{1}{2}\right)^6 = \frac{1}{64}$$

To find $P(E)$, we can use a geometric argument based on Wendel's Theorem and Euler's Formula. Imagine 6 random lines passing through the center of the planet. Each line defines a great circle on the celestial sphere. By Euler's formula for a sphere, $V - E + F = 2$, we can determine the number of regions (faces, F) these intersecting circles create.

For $N = 6$ great circles in general position:

- **Vertices (V):** Each pair of circles intersects at 2 antipodal points.
 $V = 2 \binom{6}{2} = 30$.
- **Edges (E):** Each of the 6 circles is cut by the other 5 circles at 10 points, creating 10 edges per circle.
 $E = 6 \times 10 = 60$.

Substituting into Euler's formula:

$$30 - 60 + F = 2 \implies F = 32$$

These 32 regions correspond exactly to the 32 valid configurations out of the $2^6 = 64$ possible antipodal coin-flips for the planets along those lines. Therefore:

$$P(E) = \frac{32}{64} = \frac{1}{2}$$

The conditional probability α is:

$$\alpha = P(A | E) = \frac{P(A)}{P(E)} = \frac{1/64}{1/2} = \frac{1}{32}$$

Computing β

Building a tower expands the viewing surface distance by r . On a planet of radius R , this tilts the horizon downwards by an angle $\theta = \frac{r}{R}$. The observer's visible sky expands from a hemisphere to a spherical cap with an angular radius of $\frac{\pi}{2} + \theta$.

The probability p that a single uniformly distributed planet falls within this new spherical cap is given by the ratio of the cap's surface area to the total surface area of the celestial sphere:

$$p = \frac{2\pi(1 - \cos(\frac{\pi}{2} + \theta))}{4\pi} = \frac{1 + \sin \theta}{2}$$

Let A' be the new event that all 6 planets fall into this expanded sky:

$$P(A') = \left(\frac{1 + \sin \theta}{2} \right)^6$$

We want the new conditional probability $P(A' | E) = \frac{P(A')}{P(E)}$. (For small θ , the probability that the planets fit into the expanded cap but not into any exact hemisphere scales as θ^2 and is negligible for the linear approximation).

Using the small-angle approximation $\sin \theta \approx \theta$, we can linearly approximate this conditional probability using the first-order binomial expansion:

$$P(A' | E) \approx 2 \left(\frac{1 + \theta}{2} \right)^6 = \frac{1}{32}(1 + \theta)^6 \approx \frac{1}{32}(1 + 6\theta) = \frac{1}{32} + \frac{3}{16}\theta$$

Substituting $\theta = \frac{r}{R}$ back into the equation, we get the new probability:

$$\text{New Probability} \approx \frac{1}{32} + \frac{3}{16} \left(\frac{r}{R} \right)$$

Thus, the linear coefficient is:

$$\beta = \frac{3}{16}$$

Numerical Verification

To verify the exact values of α and β , we can implement a Monte Carlo simulation.

1. **Point Generation:** We generate N random planetary systems by sampling 6 points uniformly on a 3D unit sphere (achieved by drawing from a standard 3D Gaussian distribution and normalizing the vectors).
2. **Checking Specific Hemispheres (A and A'):** The observer's specific sky is arbitrarily defined as the northern hemisphere ($z > 0$). To simulate the tower extension, we check the relaxed condition $z > -\sin \theta$ for a sufficiently small angle θ (e.g., 10^{-3}).
3. **Checking Any Hemisphere (E):** To efficiently determine if all 6 planets share *some* hemisphere without checking infinite planes, we use a geometric reduction. If a bounding hemisphere exists, its boundary (a great circle) can be continuously tilted until it intersects at least two of the planet points. Thus, we only need to test the $\binom{6}{2} = 15$ planes spanned by pairs of planets. By computing the cross-product of every pair, we generate 15 normal vectors. If all 6 planets yield a dot product of the same sign with any of these 15 normals, they lie within a shared hemisphere.
4. **Probability Calculation:** The simulated conditional probability α is the ratio of configurations satisfying Event A to those satisfying Event E . We estimate β by taking the finite difference between the simulated conditional probabilities of A' and A , divided by θ .

Simulations running 20 millions of configurations with $\theta = 0.01$ converge to the theoretical values $\alpha = \frac{1}{32}$ and $\beta = \frac{3}{16}$, with relative errors of $e_\alpha \approx 0.3\%$ and $e_\beta \approx 2.5\%$.

Conclusion

The exact values for the constants are:

$$\alpha = \frac{1}{32}, \quad \beta = \frac{3}{16}$$